Ahmedabad University

Milestone 3

Cross-Environment and SNR Robustness in Wi-Fi CSI-Based Human Activity Recognition: A Benchmark Study and Lightweight Domain-Shift Analysis

Category

AI Algorithms + Benchmark Dataset

Group 7

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Repository

https://github.com/Pranavv26/wifi-csi-domain-robustness

Table 1. Where each Milestone 3 requirement is addressed.

Requirement	Section
Final model and experimental configuration	2
Reproducible evaluation pipeline and protocol	3
Comparison with reproduced baselines and published results	4
Ablation study	5
Error analysis (failure cases, limitations, explanations)	6, 9
Revised SOTA positioning	8

Abstract

Wi-Fi channel state information (CSI) models for human activity recognition (HAR) lose accuracy in rooms and with people they were not trained on. This project is a reproduction-oriented, controlled evaluation of DATTA [1], which combines domain-adversarial training, test-time adaptation and random weight resetting on a lightweight WiFlexFormer backbone, on the Widar3.0-G6D benchmark [3]. In Milestone 3 we completed the pipeline planned in M1 and M2: we split the held-out data into a Val_TTA subset (3,417 samples) and a final Test set (30,749 samples), trained five configurations with three independent seeds on three different laptops, and analysed the results per gesture class and per unseen user. Our W + augmentation result (48.29 ± 0.11 macro-F1) lies within the spread the authors report for the same configuration (49.32 ± 4.45). Under our budget of 8 CPU epochs, however, domain-adversarial training, test-time adaptation and the full DATTA pipeline did not improve on it (full DATTA 46.59 ± 1.35, against 68.13 published). Adaptation helped on a short stream but hurt on the long Test stream, and random weight resetting recovered part of that loss in all three seeds. We report these outcomes as evidence and make no state-of-the-art claim.

1. Introduction and Scope

CSI captures the multipath channel between a transmitter and a receiver, so the same gesture produces different CSI in another room or for another person, and models trained in one setting often degrade in another. M1 [2] scoped this problem and selected DATTA [1] as the reference method, with a staged reproduction plan. M2 [5] built the Widar3.0-G6D data pipeline and trained the vanilla WiFlexFormer baseline W (one run, 15 epochs, CPU; 45.96% macro-F1 on the merged 34,166-sample held-out set). It left the remaining work for M3: separate Val_TTA from Test, then add augmentation, domain-adversarial training (DAT), test-time adaptation (TTA) and random weight resetting. This report documents that work and is organised around the Milestone 3 requirements (Table 1). The project is a reproduction and component-wise study of an existing method, not a new algorithm.

Our contributions are: (i) a controlled single-codebase pipeline with a corrected Val_TTA/Test protocol; (ii) nine training runs (three seeds × three trained configurations) on three machines, plus six adaptation runs; (iii) a five-row ablation with per-seed effects and TTA tuning on Val_TTA only; and (iv) a per-class and per-user error analysis, including a short-versus-long stream finding for test-time adaptation.

2. Final Method and Experimental Configuration

2.1 Model and components

The backbone is WiFlexFormer [4]: a convolutional stem, Gaussian positional encoding and a transformer encoder with a class token, followed by a linear activity classifier. It has 41,977

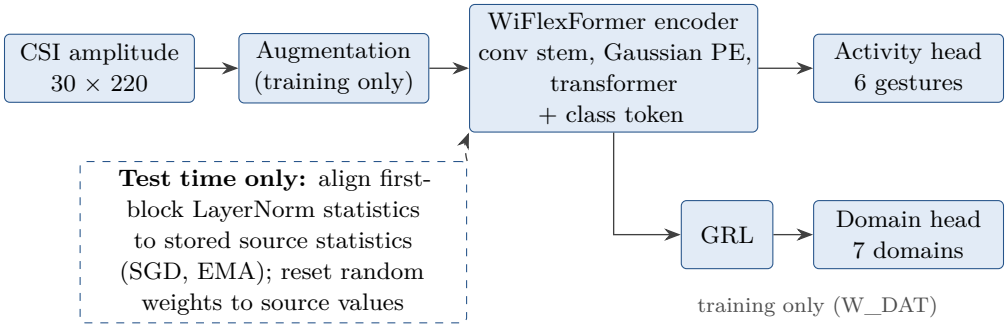


Figure 1. Pipeline of the final configuration (full DATTA). Augmentation and the gradient reversal layer (GRL) with its domain head are used only during training; statistic alignment and random weight resetting are used only at test time. W uses only the path from input to activity head.

Table 2. Components added in each ablation row.

Configuration	Augmentation	DAT (GRL + CCC)	TTA	Reset
W				
W + aug	✓			
W_DAT + aug	✓	✓		
W_DAT + aug + TTA	✓	✓	✓	
DATTA (full)	✓	✓	✓	✓

parameters in every configuration. The input is the CSI amplitude of 30 subcarriers over 220 packets. Figure 1 shows the final pipeline, and Table 2 lists which component each ablation row adds.

Augmentation. CSI-specific perturbations applied to training data only: amplitude brightness change, pixel-wise and column-wise dropout and circular rotation (authors' configuration; horizontal flipping is disabled in it).

DAT. A domain discriminator behind a gradient reversal layer pushes the encoder towards features that do not reveal the training domain (7 domains). The confidence-control constraint (CCC) is an additional loss term on the activity predictions; in the released code it is the mean prediction entropy (see Section 9).

TTA. At test time the model adapts online to the unlabelled stream. Its parameters are updated by SGD so that the feature statistics of the LayerNorm layers in the first encoder block match statistics stored from the training set, using an exponential moving average (EMA) of the target statistics.

Random weight resetting. After every adaptation step each weight is independently reset to its source value with a small probability, which limits drift and forgetting.

2.2 Attribution

DAT, the confidence-control loss, TTA and random weight resetting are the authors' released implementation [1] (`github.com/StrohmayerJ/DATTA`). Our contributions are the controlled pipeline and Val_TTA/Test protocol, CPU fixes to the TTA code, the multi-seed, multi-machine runs, the TTA tuning, the re-evaluation scripts, the tables and figures, and the error analysis.

2.3 Configuration

Table 3 gives the settings. All three trained configurations (W, W + aug, W_DAT + aug) use identical training settings; the TTA settings apply to the two adaptation rows.

Table 3. Experimental configuration.

Item	Setting
Training	8 epochs, batch size 8, AdamW (learning rate 5×10^{-5}, weight decay 10^{-3}), cosine annealing with warm restarts, CPU only, seeds 1, 2, 3
Source split	80/20 train/validation (19,586 / 4,896), fixed seed 42; checkpoint = lowest source-validation loss
Augmentation	Amplitude brightness (± 0.01), pixel-wise and column-wise dropout (rate 10^{-3}) and circular rotation (range 0.5), each applied with probability 0.5
DAT	GRL scale 8 with the authors' ramp-up, domain-loss weight 0.3, confidence-control weight 0.2
TTA	SGD, learning rate 10^{-4}, EMA momentum 0.9, batch size 1, fixed order, one pass over the stream; chosen on Val_TTA (Section 5.2)
Resetting	Per-weight reset probability 10^{-4} after every sample (0 for the TTA-only row)

Table 4. Data subsets and their roles in M3.

Subset	Domains	Samples	Role
Source training	rooms 2–3 (7 domains)	19,586	Fit model weights
Source validation	rooms 2–3 (7 domains)	4,896	Checkpoint selection (lowest loss)
Val_TTA	room 1 (9 users)	3,417	Tune TTA settings only
Test	room 1 (9 users)	30,749	All reported results

3. Experimental Setup and Evaluation Protocol

3.1 Dataset and splits

We use Widar3.0-G6D [1, 3], the six-gesture subset of Widar3.0 with class order Push&Pull, Sweep, Clap, Slide, Draw-circle and Draw-zigzag. A domain is one room–participant combination. Training uses 24,482 samples from 7 domains (rooms 2 and 3); the held-out set has 34,166 samples from 9 domains (room 1, users 5, 10, 11, 12, 13, 14, 15, 16 and 17). Training and held-out data share no domain, so every held-out sample comes from an unseen room and an unseen user.

Change from M2. In M2 the 34,166 held-out samples were evaluated as one set. For M3 we split them 10/90 with a fixed seed (42) into **Val_TTA** (3,417 samples), used only to tune TTA settings, and **Test** (30,749 samples), used for every reported number (Table 4). These sizes match the paper's Val_TTA and Test subsets [1]. Val_TTA is a random part of the same nine unseen domains, and its labels are used only to score tuning runs. No setting was chosen on Test.

3.2 Metrics

The main metric is macro-F1, computed as $2PR/(P+R)$ from macro-averaged precision P and recall R, which is the convention of the authors' code. We also report accuracy, precision, recall, per-class recall, row-normalised confusion matrices and per-user macro-F1. Means and standard deviations are taken over the three seeds (population standard deviation).

3.3 Three seeds on three machines

Each seed was trained end-to-end for all three trained configurations, giving nine training runs on three laptops (Table 5). The code, flags and data caches (copied from the same source files) were the same, so the seeds also act as a cross-hardware reproducibility check. TTA and full DATTA

Table 5. Training hardware per seed (all CPU).

Seed	Trained by	Machine	Time per epoch
1	Pranav Jain	Windows laptop, AMD Ryzen 5 5500U	about 30–40 min
2	Paarth Jawaharani	Apple M2 MacBook	about 5 min
3	Kritika Lunkad	Apple M2 MacBook	about 4–5 min

Table 6. Test results (30,749 samples, %): our mean ± std over three seeds versus the published values [1] (3-run means with std where reported). Published values are the authors' results, not ours.

Configuration	Macro-F1	Acc.	Prec.	Rec.	Paper F1	Ours − paper
W	47.43 ± 0.44	44.72	47.63	47.24	40.62 ± 2.93	+6.81
W + aug	48.29 ± 0.11	45.63	48.51	48.07	49.32 ± 4.45	−1.03
W_DAT + aug	46.81 ± 0.83	43.94	46.87	46.77	65.66 ± 1.85	−18.85
W_DAT + aug + TTA	45.13 ± 1.47	42.63	46.18	44.16	–	–
DATTA (full)	46.59 ± 1.35	44.33	46.91	46.29	68.13	−21.54

were then run on each seed's own W_DAT checkpoint, giving six further Test runs, all executed on the seed-1 laptop. Best epochs (out of 8) were 6/7/8 for W, 8/7/8 for W + aug and 6/7/8 for W_DAT + aug, for seeds 1/2/3.

All Test numbers for W, W + aug and W_DAT + aug come from re-evaluating the saved checkpoints on the seed-1 laptop, so every row uses the same data copy and numerics. Seeds 1 and 2 reproduce their training-time results exactly. For seed 3 the re-evaluated numbers are 0.2–0.3 F1 higher than its training-time results (46.81 / 48.08 / 46.75 for W / W + aug / W_DAT + aug); the seed was trained and first evaluated on a different laptop, and we have not identified the cause (a difference in the data copy or in numerics is possible).

3.4 Reproducibility

The scripts, per-run result files (confusion matrix and per-sample predictions) and logs are in the `M3_g7_aialgorithm_wifisensing` folder of the project repository (`Code/SRC` and `Results`). Every table and figure in this report is generated from those files by `make_report_assets.py`. The dataset class, model code and augmentation configuration come from the authors' repository [1].

4. Results and Comparison with Published Values

Table 6 compares our mean results with the published values and Table 7 lists every seed. **Comparability:** the authors trained much longer, on different hardware and with their own hyperparameters; we trained for 8 epochs on CPU. The comparison is therefore "wherever comparable", not like-for-like. Our M2 number (45.96) used a merged held-out set and different settings, so it is context only and not a row here.

- **W + aug is reproduced closely.** Our 48.29 ± 0.11 is about 1 point below the published 49.32 and well inside its reported spread of ±4.45.
- **Our plain W is stronger than the published W** (47.43 against 40.62). We report this as a difference, not as confirmation of the paper.
- **The DAT and full DATTA gains are not reproduced.** W_DAT + aug (46.81) and full DATTA (46.59) stay close to the baselines and far below the published 65.66 and 68.13.

Table 7. Test results for each seed (%).

	Macro-F1			Accuracy		
Configuration	**Seed 1**	**Seed 2**	**Seed 3**	**Seed 1**	**Seed 2**	**Seed 3**
W	48.05	47.20	47.05	46.18	44.09	43.88
W + aug	48.15	48.42	48.28	46.04	45.37	45.48
W_DAT + aug	47.70	45.71	47.03	45.66	42.15	44.02
W_DAT + aug + TTA	44.23	43.95	47.19	41.41	42.05	44.43
DATTA (full)	47.10	44.75	47.93	44.61	42.96	45.41

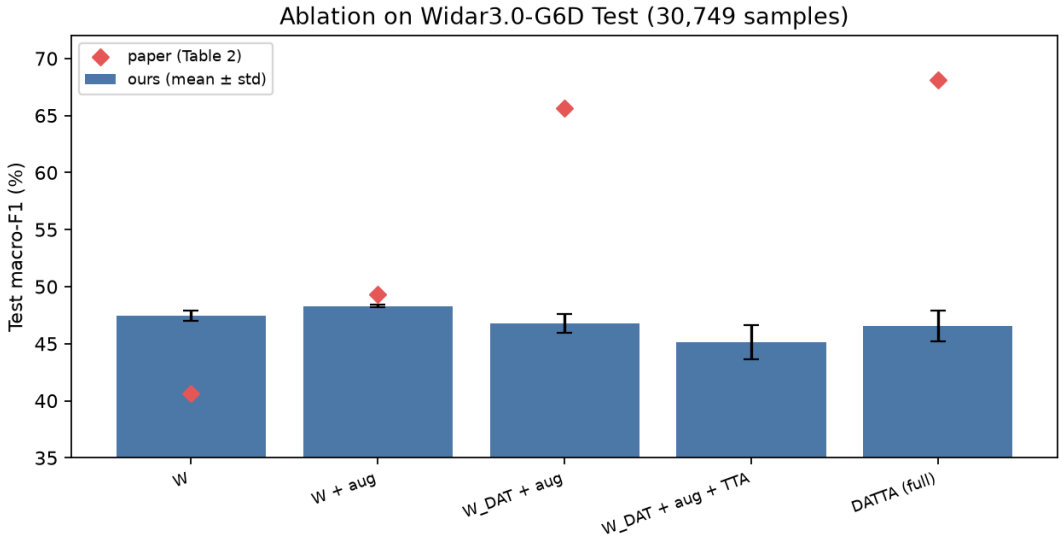


Figure 2. Ablation on Test: our mean macro-F1 with standard deviation over three seeds (bars) and the published values [1] (red markers; none is used for the TTA-only row).

5. Ablation Study

5.1 Effect of each component

Figure 2 shows the five configurations and Table 8 gives the effect of adding each component, computed seed by seed so that the consistency of each effect is visible.

Augmentation raised macro-F1 in all three seeds (mean +0.85) and gave the lowest variance of any configuration. **DAT** lowered F1 relative to W + aug in all three seeds (mean −1.47). **TTA alone** lowered it further in two of three seeds (mean −1.69), whereas **random weight resetting** improved on TTA in all three seeds (mean +1.47). Full DATTA ends 0.22 below the unadapted W_DAT + aug model and 1.70 below W + aug. Differences below about one point, such as W

Table 8. Change in Test macro-F1 caused by each component, per seed (percentage points).

Effect	**Seed 1**	**Seed 2**	**Seed 3**	**Mean**
CSI augmentation (W + aug − W)	+0.10	+1.22	+1.23	+0.85
Domain-adversarial training (W_DAT − W + aug)	−0.45	−2.71	−1.25	−1.47
Test-time adaptation (TTA − W_DAT)	−3.47	−1.76	+0.16	−1.69
Random weight resetting (DATTA − TTA)	+2.87	+0.80	+0.74	+1.47
TTA + resetting together (DATTA − W_DAT)	−0.60	−0.96	+0.90	−0.22

Table 9. TTA tuning on Val_TTA (seed-1 W_DAT checkpoint, 3,417 samples).

Setting	Val_TTA macro-F1	Δ vs. no adaptation
No adaptation (learning rate 0)	45.49	–
Paper default: momentum 0.55, lr 10^{-4}	45.91	+0.42
Momentum 0.55, lr 10^{-3}	45.18	−0.31
Momentum 0.55, lr 10^{-2}	36.45	−9.04
Momentum 0.9, lr 10^{-4} (chosen)	**46.34**	**+0.85**
Momentum 0.99, lr 10^{-4}	46.36	+0.87
Momentum 0.9, lr 10^{-4}, reset 10^{-4}	46.17	+0.68
Momentum 0.9, lr 10^{-4}, reset 10^{-3}	45.70	+0.21

Table 10. Per-class recall on Test (%, mean over three seeds).

Class	W	W + aug	W_DAT	+ TTA	DATTA	DATTA − W
Push&Pull	38.54	35.23	41.71	54.70	54.22	+15.68
Sweep	70.26	71.80	73.20	77.89	78.20	+7.94
Clap	73.99	71.90	76.39	49.55	59.65	−14.34
Slide	43.18	45.54	39.06	15.84	19.20	−23.97
Draw-circle	20.37	22.99	24.37	16.18	15.08	−5.29
Draw-zigzag	37.12	40.94	25.89	50.80	51.39	+14.27

versus W + aug, should not be over-interpreted with three seeds; the consistent sign across seeds is the stronger evidence.

5.2 TTA tuning on Val_TTA

TTA settings were tuned on Val_TTA only, using the seed-1 W_DAT checkpoint (Table 9); the adaptation stream uses batch size 1, a fixed order and one pass. Momentum 0.99 tied with 0.9 (within 0.02), and 0.9 was kept because the reset sweep had used it. The final settings were momentum 0.9, learning rate 10^{-4} and no reset for the TTA row, and the same plus reset probability 10^{-4} (the paper's default) for full DATTA. We kept the default reset rate because a 3,417-sample stream is too short to show forgetting.

5.3 Short versus long streams

The effect of the chosen settings on the two streams, relative to the unadapted model, differs in sign. On Val_TTA (seed 1), TTA improved macro-F1 by 0.85 and DATTA by 0.68. On the 30,749-sample Test stream the same settings changed it by −3.47 (TTA) and −0.60 (DATTA) for seed 1, and by −1.69 and −0.22 on average over the three seeds. This pattern is consistent with gradual drift or forgetting over a long unlabelled stream, which is the problem random resetting is designed to address [1], and resetting did recover part of the loss. We did not run a controlled experiment that varies only the stream length, so the drift explanation remains an interpretation. Even with resetting, adaptation did not beat the unadapted W_DAT + aug model.

6. Error Analysis

6.1 Per-class behaviour

Table 10 gives per-class recall, and Figure 3 shows the confusion matrices of W and full DATTA.

Sweep and Clap are recognised best and Draw-circle is the hardest class for W, W + aug and W_DAT

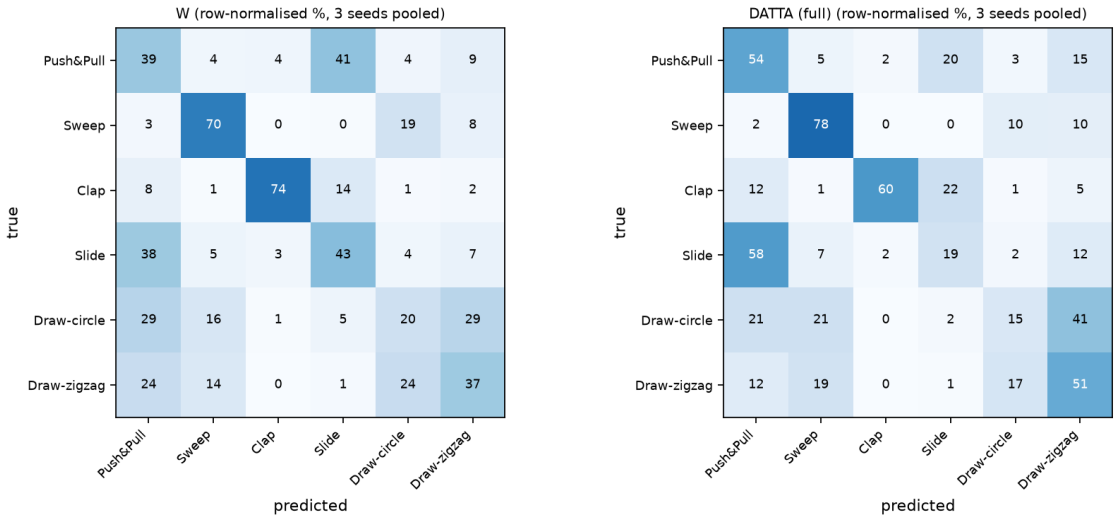


Figure 3. Row-normalised confusion matrices on Test (%, three seeds pooled): W (left) and full DATTA (right).

+ aug. In W the main confusions (percentage of the true class, seeds pooled) are Push&Pull→Slide (40.7), Slide→Push&Pull (37.6), Draw-circle→Push&Pull (29.4), Draw-circle→Draw-zigzag (29.0), Draw-zigzag→Draw-circle (23.7). These pairs involve similar overall hand motion (pushing and pulling versus sliding, and the two drawing gestures), which is a plausible explanation that we have not tested.

TTA changes the shape of the predictions rather than improving them evenly. Push&Pull, Draw-zigzag and Sweep gain recall, while Slide and Clap collapse. In full DATTA the largest confusions become Slide→Push&Pull (57.9), Draw-circle→Draw-zigzag (41.3), Clap→Slide (21.6), Draw-circle→Sweep (20.9), Draw-circle→Push&Pull (20.6), Push&Pull→Slide (20.1). Our reading is that unsupervised statistic alignment shifts the class bias of the model; we did not test this further.

6.2 Per-user behaviour

Figure 4 shows macro-F1 for each unseen user. User 17 is the clearest failure case: 24.3% for W and 20.7% for DATTA, against roughly 43–61% for the other users with W. Because no configuration helps this user, the cause probably lies in the data rather than in any one component, but we have not investigated it. User 12 is the only user clearly helped by adaptation (43.0% for W, 46.2% for DATTA), while users 13, 14 and 16, among the best for the baselines, lose 2.5 to 6.3 points under DATTA relative to W + aug.

6.3 Training behaviour

Figure 5 shows source-validation macro-F1 and training loss. W and W + aug reach most of their final value within three epochs (W: 45.2% after epoch 1, 49.7% after epoch 3 and 50.0% after epoch 8). W_DAT + aug stays near 46.1% for four epochs, rises to 47.4% at epoch 5 and ends at 47.6%, so it is also below W on in-domain validation data. The curves are flat over the last epochs, so our runs give no evidence on whether longer training would close the gap to the published DAT result; we list training length as an untested possibility (Section 9).

7. Complexity

All configurations have 41,977 parameters; the domain head is used only during training. Table 11 gives the inference cost on the seed-1 laptop CPU. The streaming rows include the adaptation step and data loading and were measured while other applications were running, so they are approximate;

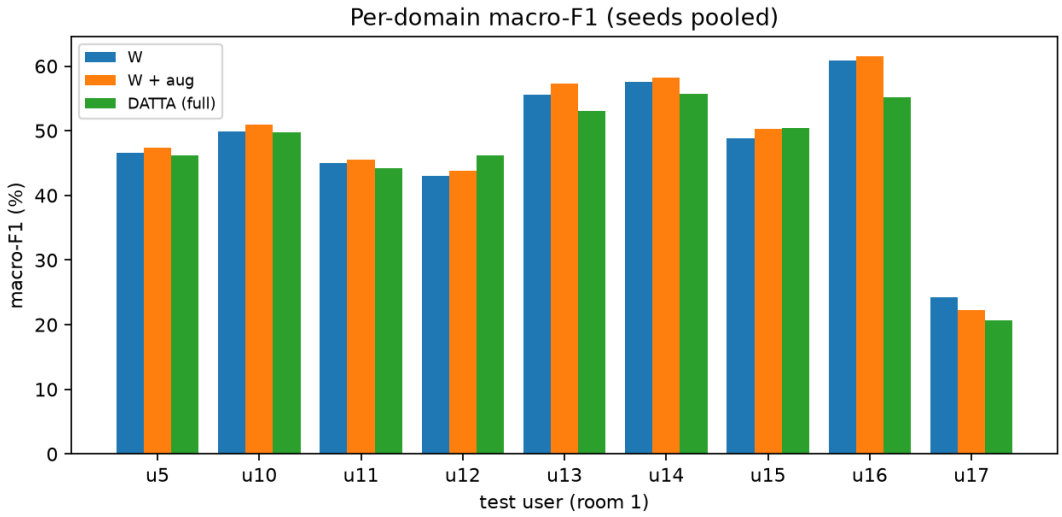


Figure 4. Test macro-F1 per unseen user (seeds pooled) for W, W + aug and full DATTA.

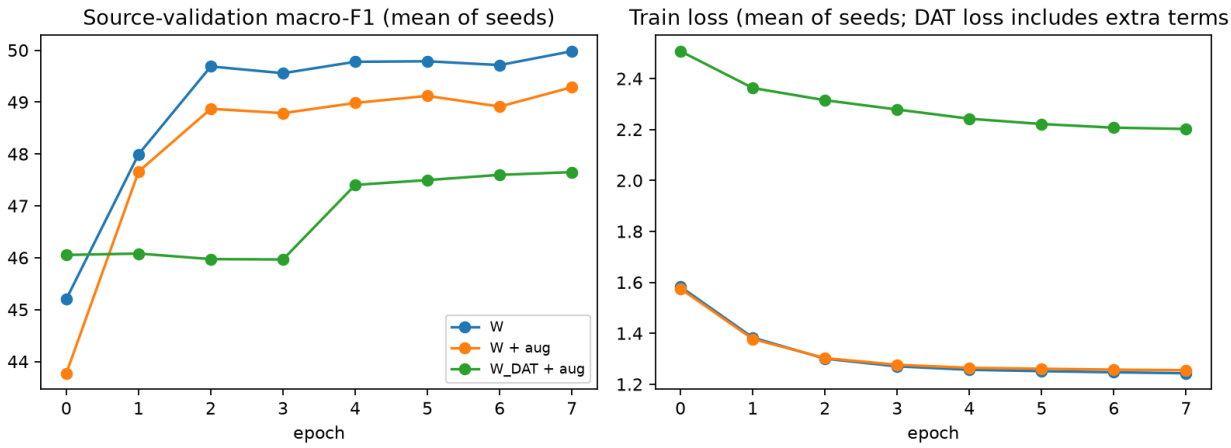


Figure 5. Source-validation macro-F1 (left) and training loss (right) per epoch, mean of three seeds. The x-axis starts at 0, so "epoch n" in the text is $x = n - 1$. The DAT loss includes extra terms, so raw losses are not comparable across configurations.

even so, a sample is processed in well under 0.1 s with adaptation.

8. Revised SOTA Positioning

DATTA [1] remains the most directly relevant published reference for cross-domain HAR on Widar3.0-G6D: it addresses the same unseen-room, unseen-user setting with a lightweight model and online adaptation. Our independent, compute-limited reproduction confirms part of its story and leaves the rest unconfirmed (Table 12). We make no state-of-the-art claim, and we do not

Table 11. Inference cost per sample on the seed-1 laptop CPU (AMD Ryzen 5 5500U).

Mode	ms per sample
Plain inference, batch size 128	5.4
Plain inference, batch size 1	9.5
Online TTA, batch size 1 (stream wall time)	about 34–41
Online TTA + random resetting, batch size 1 (stream wall time)	about 55–62

Table 12. Published claims against our evidence.

Published claim [1]	Our evidence (3 seeds)	Assessment
Plain W reaches 40.62 F1	47.43 ± 0.44	Not reproduced; our baseline is stronger
Augmentation raises F1 (W + aug 49.32)	48.29 ± 0.11; gain in 3 of 3 seeds	Supported; within the published spread
DAT adds a large gain (65.66)	46.81 ± 0.83; below W + aug in 3 of 3 seeds	Not reproduced at 8 CPU epochs
Full DATTA reaches 68.13	46.59 ± 1.35	Not reproduced at 8 CPU epochs
Random resetting limits forgetting in TTA	+1.47 over TTA; gain in 3 of 3 seeds	Qualitatively supported

claim the paper is wrong.

The open questions our results identify are the training length needed for DAT to pay off, and how TTA settings should be chosen when the deployment stream is much longer than the tuning stream.

9. Limitations and Plan for M4

9.1 Limitations

- **Training budget.** We trained for 8 epochs on CPU. The authors' released training script defaults to 4,000 epochs; we did not find an epoch count in the paper's main text. Training length is a plausible but untested explanation for the missing DAT gain. As indirect evidence, earlier 30-epoch runs of W and W + aug on an older code branch (44.60 ± 0.62 and 48.53 ± 0.28; not directly comparable) did not improve over 8 epochs, so the baselines plateau early. Whether DAT behaves differently with much longer training is unknown.
- **CCC implementation.** The paper describes the confidence-control constraint as penalising over-confident predictions [1]. In the released code we ran, the CCC term is the mean prediction entropy added to the loss with weight 0.2, and minimising entropy favours confident predictions. We ran the code unchanged and did not test whether this difference affects the DAT result; it is a further candidate explanation, alongside training length.
- **Seeds and settings.** Three seeds and one training hyperparameter setting; with three seeds the standard deviations are themselves noisy.
- **TTA tuning.** TTA was tuned on one seed and on a short stream, and the reset rate was not tuned. The chosen momentum may not suit the much longer Test stream; we did not test other settings on Test.
- **Checkpoint selection** used source-validation loss, which may not track target-domain performance.
- **Hardware.** Seeds ran on different machines, which may cause small numeric differences (the seed-3 offset in Section 3); streaming timings were measured with other applications running.
- **Single dataset.** The paper also evaluates on 3DO and MM-Fi; we used Widar3.0-G6D only.

9.2 Plan for M4 (not yet done)

1. Train W_DAT for much longer (tens to hundreds of epochs) to test the training-length hypothesis, and check the CCC term against its description in the paper.
2. Tune TTA momentum and the reset rate on a longer stream and on all seeds.
3. Run on a GPU if available, with a compute-matched comparison to the paper's schedule.

References

[1] J. Strohmayer, R. Sterzinger, M. Wödlinger, and M. Kampel, "DATTA: Domain-Adversarial Test-Time Adaptation for Cross-Domain WiFi-Based Human Activity Recognition," in *Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV)*, 2026. Code: https://github.com/StrohmayerJ/DATTA

[2] P. Jain, P. Jawaharani, and K. Lunkad, "Milestone 1: Cross-Environment and SNR Robustness in Wi-Fi CSI-Based Human Activity Recognition," ECE 310 Wireless Communications, Ahmedabad University, Group 7, 2026.

[3] Y. Zhang *et al.*, "Widar3.0: Zero-Effort Cross-Domain Gesture Recognition with Wi-Fi," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 8671–8688, 2022.

[4] J. Strohmayer, M. Wödlinger, and M. Kampel, "WiFlexFormer: Efficient WiFi-Based Person-Centric Sensing," 2024.

[5] P. Jain, P. Jawaharani, and K. Lunkad, "Milestone 2: Baseline reproduction (vanilla WiFlexFormer on Widar3.0-G6D)," ECE 310 Wireless Communications, Ahmedabad University, Group 7, 2026.